Baher Mikhael Habib Azer

Junior Data Scientist | Python & Machine Learning

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linkedin.com/in/baher-data-analyst | github.com/baherend | Portfolio

PROFESSIONAL SUMMARY

Junior Data Scientist and Data Analyst with a Behavioral Psychology background and hands-on experience in a telecommunications environment. Skilled in Python (Pandas, NumPy, scikit-learn, Matplotlib, Plotly), SQL, and exploratory data analysis (EDA), with machine learning and deep learning projects in PyTorch and TensorFlow/Keras, computer vision (OpenCV), and LLM-based systems (RAG). Experienced in collecting, cleaning, and organizing data, building Python scripts for data processing and visualization, and documenting reusable workflows. Self-driven learner eager to apply data science to RF network performance data and network optimization.

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Plotly), SQL, R (basic), Jupyter Notebook, Git/GitHub

Machine Learning & AI: scikit-learn (regression, classification, clustering, model evaluation), PyTorch, TensorFlow/Keras, CNN, LSTM, Computer Vision (OpenCV), Large Language Models (LLMs), RAG, NLP (Transformers), Gradient Descent-based Optimization (Adam/AdamW)

Data Analysis: Exploratory Data Analysis (EDA), Data Cleaning and Wrangling, Feature Engineering, Statistical Analysis, Trend and Performance Analysis, Data Visualization, Data Storytelling, ETL Pipelines

BI & Tools: Power BI (DAX, Data Modeling), Advanced Excel (Power Query, Pivot Tables), SQL Server (SSMS), Streamlit, Gradio, FastAPI

Professional: Analytical thinking, attention to detail, self-learning, problem solving, clear communication, teamwork, documentation

PROJECTS

Smart-City CCTV Violence Detection | Computer Vision, PyTorch | github.com/baherend/Smart-City-CCTV-Violence-Detection

- Built a ResNet18 + LSTM video classifier (PyTorch, OpenCV) for Normal / Violence / Weaponized, reaching 99.8% test accuracy on the dataset's pre-segmented 1-second clips; evaluated on full-length clips (67.1%) to assess generalization limits.
- Trained with the Adam optimizer and learning-rate scheduling, evaluated with scikit-learn (classification report, confusion matrix), and delivered an interactive Plotly probability timeline in a Gradio/Streamlit app.

Football Analytics RAG System, FIFA World Cup 2022 | LLM, NLP, Python | github.com/baherend/Football-Analytics-RAG-FULL

- Built a Retrieval-Augmented Generation (RAG) system over StatsBomb open data (64 matches) to answer natural-language questions about matches, players, and teams across 2,835 indexed documents.
- Combined BM25 lexical search with dense embeddings via Reciprocal Rank Fusion, plus a query router; built a modular Python data pipeline (extraction, preprocessing, chunking) with automated tests, a CLI, and a Streamlit app (ChromaDB, OpenRouter LLM).

E-Commerce Sales Analysis and Customer Segmentation | Python, EDA | github.com/baherend/E-Commerce-Sales-Analysis

- Cleaned and analyzed ~542K retail transactions (Pandas, NumPy), removing 5,268 duplicates and isolating 9,071 returns to calculate true net revenue.
- Performed EDA and visualization (Matplotlib, Seaborn) to identify peak-order hours and bulk-buyer revenue spikes, segmented customers into 3 tiers by purchase frequency and value, and ran Market Basket Analysis on co-purchased products.

Audio Doctor: Call Review Prototype | Audio Signal Analysis, PyTorch | github.com/baherend/audio-review

- Built a Streamlit prototype that lets call-center team leaders review recorded Arabic calls from acoustic and timing evidence only: stereo channel validation, RMS-based activity gating, windowed vocal-activation classification (Wav2Vec2), and heuristic hold and dead-air detection.
- Aggregated results into an exportable call review (JSON/PDF) with unit tests (pytest); pitched the concept to call-center companies.

Interview IQ: Multimodal Interview Intelligence System | Graduation Project | github.com/baherend/InterviewIQ

- Developed an AI platform fusing speech, text, and visual signals into unified performance scores through separate NLP, computer vision, and audio pipelines (PyTorch, OpenCV, Librosa, Transformers), with LoRA fine-tuning of transformer models and a FastAPI backend.

End-to-End Data Analysis and Visualization Project | SQL Server, Excel

- Designed a relational database in SQL Server (SSMS) and built an interactive Excel dashboard to visualize KPIs; published insights on LinkedIn.

RELEVANT EXPERIENCE

Technical Support Specialist (Voice Account) | WE Communications (Telecom) *2023 – 2024*

- Resolved 40+ technical issues daily on telecom voice services with a 95% CSAT, and reported recurring technical trends from customer data to improve response time.

Psychologist and Behavioral Specialist | Bashayer Al Khair Foundation *Feb 2025*

- Analyzed behavioral data patterns for 15+ cases to identify trends and optimize treatment efficacy by 30%, using quantitative metrics to track progress.

ADDITIONAL EXPERIENCE

- **Psychological Lecturer**, EIME Center (Summer 2019): delivered 10+ lectures to audiences of 50+, driving a 40% increase in workshop engagement.
- **Pharmacy Assistant**, Multiple Pharmacies (2017 – 2022): managed inventory data for 200+ products, reducing stock discrepancies by 15% through data-driven tracking.

EDUCATION

Digital Pioneers (Digilians) Scholarship | MCIT and The Egyptian Military Academy: Data Analysis and Applied Artificial Intelligence *Dec 2025 – Present (Expected Nov 2026)*

Bachelor of Arts in Psychology | Alexandria University *Graduated 2019*

Cognitive Behavioral Therapy (CBT) Diploma | Wataneya Institute *2019*

CERTIFICATIONS

- Microsoft Certified: Power BI Data Analyst Associate (PL-300), Microsoft
- Microsoft Certified: Azure Data Fundamentals (DP-900), Microsoft
- Google Data Analytics Professional Certificate, Google (Coursera)
- Machine Learning with Python (scikit-learn), IBM, Coursera
- Introduction to Deep Learning and Neural Networks with Keras, IBM
- Data Analysis with R Programming, Google, Future Skills Academy
- Preparing Data for Analysis with Microsoft Excel, Microsoft, Future Skills Academy (May 2026)

In progress: Microsoft Power BI Data Analyst and AI Engineer specializations, Future Skills Academy.

LANGUAGES

Arabic: Native | English: Professional Working Proficiency (B1) | German: Elementary (A2, ongoing)